01 · Who should we email? — customer-level uplift (Anchor A)

July 12, 2026

The business decision. We have a discount offer and a customer list; contacting one customer costs **€8 all-in** (set with the simulation below — the bar a customer's uplift must clear). Emailing *everyone* burns margin on people who would have bought anyway, and a discount email can even annoy some loyal customers into unsubscribing. Emailing *no one* leaves money on the table. So the question is not "does the email work on average?" but **"for which individual customers does sending the email actually change what they do?"**

0.0.1 The one idea this whole notebook rests on

For each customer there are two possible futures: the spend we'd see **if we email them**, and the spend we'd see **if we don't**. Statisticians call these two futures *potential outcomes* and write them $Y(1)$ (emailed) and $Y(0)$ (not emailed). The thing we care about is the **difference** — how much the email *changes* that customer's spend:

$$\tau_i \;=\; Y_i(1) - Y_i(0). \qquad \text{(the customer's individual treatment effect, in €)}$$

We use the Greek letter **tau ("tau")** for this effect throughout. When it depends on the customer's characteristics x (recency, engagement, …) we write it as a function $\tau(x)$.

The catch — the **"fundamental problem of causal inference"** — is that for any one customer we only ever get to see *one* of the two futures: we either emailed them or we didn't, never both. So τ_i is never directly observed for anybody. Everything that follows is a disciplined way of *filling in the future we didn't see.*

0.0.2 Four kinds of customer — and why only the effect tells them apart

Sort customers by what they do in each of the two futures, and four types appear. Notice that two customers can have the *same observed behaviour* ("bought after being emailed") yet be completely different types:

	buys if emailed	**doesn't buy if emailed**
buys if not emailed	(yellow) *Sure Thing* — would buy anyway, so tau ~ 0 (the discount is pure waste)	(red) *Sleeping Dog* — the email **backfires**, tau < 0 (reminding them prompts an unsubscribe/return)

	buys if emailed	doesn't buy if emailed
doesn't buy if not emailed	(green) *Persuadable* — the email *tips them over*, tau > 0 (**the only type worth paying to reach**)	(white) *Lost Cause* — won't buy regardless, tau ~ 0

Only the **Persuadables** are worth a discount. But you cannot identify them from who bought — a Sure Thing and a Persuadable both "bought after the email." You can only separate them by estimating the *effect* tau, which is why plain response models (predicting who buys) are the wrong tool and **uplift models** (predicting whose behaviour *changes*) are the right one.

Targeting is worthwhile **only because tau varies across customers** ("heterogeneity"). If every customer had the same tau, there'd be nothing to select on — you'd rationally email everyone (if tau > cost) or no one (if tau < cost). So the entire problem reduces to: **estimate the per-customer effect $\tau(x)$ well, then turn it into a euro decision that respects how unsure we are.**

0.0.3 How this notebook is organised

It follows the repo's fixed **7-step contract**, then adds **three depths** a data scientist needs to defend the number to a skeptic:

- **Steps** — 1 question · 2 simulate a known truth · 3 identify · 4 estimate · 5 validate · 6 decide in € · 7 caveats
- **Depth A** estimator bake-off & failure modes · **Depth B** identification rigour & sensitivity · **Depth C** euro policy, value-of-information & test design

> **A note on "Bayesian."** We fit models that return not a single number but a *distribution* of plausible values for each effect — a **posterior**. In the narration we never dwell on the machinery; a posterior just lets us say *"how sure are we, and does that change what we'd do?"* A **90% credible interval** below means "given the data and model, there's a 90% chance the true value lies in this range."

```
FAST=False  N=1600  sampling={'draws': 600, 'tune': 600, 'chains': 4, 'm': 100}
```

0.1 2 · Simulate a ground truth (why we start with fake data)

Here is the uncomfortable fact that shapes how we *validate* any causal method: **on real data we can never check whether the method got the effect right**, because the true effect τ_i is never observed for anyone (we only saw one of their two futures). If we just ran a fancy model on real data and it printed "€6.10", we'd have no way to know if that's right or nonsense.

The way out — used in every notebook here — is **validation-first**: we build a *simulator* that plants a **known** effect for each customer, generate data from it, and then see whether the method **recovers the number we planted**. Only after a method passes that test on data where we know the answer do we trust it on data where we don't. (Section 6b then runs the same pipeline on a real public dataset, where we can no longer grade recovery but can still make the decision.)

What the simulator plants. Each customer gets:

- five features x — `recency` (days since last purchase), `frequency`, `monetary`, `tenure`, `engage` (an engagement score);

- a true per-customer effect $\tau(x)$ (in €), deliberately *heterogeneous* — engaged, mid-recency customers are **Persuadables** (large positive tau); habitual high-frequency but low-engagement customers are **Sleeping Dogs** (tau < 0, the email nudges them to unsubscribe); the rest sit near 0 (tau is continuous, so nobody is *exactly* zero — the Step-2 printout therefore shows only the two signed shares);
- a **selection rule** that decides who got emailed in the past. In the *observational* world we use here, marketers historically emailed the already-engaged, recently-active customers — so whether a customer was emailed is **tangled up with the kind of customer they are**. That tangling is called **confounding**, and it is exactly what makes the naive answer wrong (Step 4 and Depth A show it).

The data-generating model — exactly what `dgp.uplift_customers` implements (defaults & seed in `src/cmp/dgp.py`). Features: R = recency $\sim U(5, 330)$, F = frequency $\sim$ Poisson(5), M = monetary $\sim U(15, 180)$, tenure $\sim U(1, 60)$, E = engage $\sim$ Beta(2, 3). With $\sigma(z) = 1/(1 + e^{-z})$ the logistic function:

$$\mu_0(x) = \max\Big\{0,\ 20 + 1.4\,F + 0.25\,M + 12\,E - 0.03\,R\Big\}$$

$$\tau(x) = \underbrace{42\,e^{-\left(\frac{R-120}{70}\right)^2}\,\sigma(F-3)\,\sigma(9(E-0.28))}_{\text{Persuadables: engaged, mid-recency}} - \underbrace{24\,\sigma(9(0.22-E))\,\sigma(1.2(F-6))}_{\text{Sleeping Dogs: frequent but disengaged}}$$

$$e(x) = \sigma\Big(1.8\,(E-0.4) + 1.2\,\tfrac{150-R}{150} + 0.15\,(F-5)\Big), \qquad T \sim \text{Bernoulli}(e(x))$$

$$Y = \mu_0(x) + \tau(x)\,T + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0,\,8^2).$$

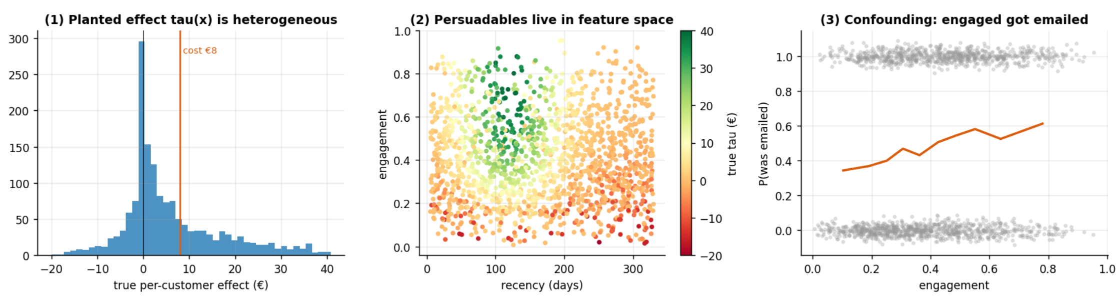
Line 1 is the baseline spend if not emailed; line 3 is the historical (observational) assignment rule. Three things to read off the equations. **(i) The confounding is visible**: engagement and recency raise the emailing propensity $e(x)$ *and* raise spend through $\mu_0(x)$ — that double role is what breaks the naive comparison. In the *randomized* regime the third line is replaced by $e(x) = \frac{1}{2}$ and the confounding vanishes. **(ii) `tenure` is a red herring**: it is drawn but enters neither μ_0 nor τ, so a good estimator must learn to ignore it. **(iii) Depth B's hidden confounder** adds an unobserved $U \sim \mathcal{N}(0, 1)$ to both the propensity logit ($+\,a_T U$) and the outcome ($+\,b_Y U$) — the two knobs the sensitivity analysis sweeps.

```
True ATE (average of tau over everyone) = €5.95
Share historically emailed              = 48%

Share of each uplift type in the base:
persuadable     0.7
sleeping dog    0.3
  (of the base, profitably persuadable at cost €8 - tau > cost: 30%)

      recency  frequency   monetary     tenure    engage    T          y  \
0  208.156027        4.0  84.060202  44.320568  0.545720  1.0  45.878520
1  296.594485        3.0  48.303531  32.765013  0.518337  1.0  32.005618
2  257.097849        5.0  46.447918  51.242745  0.921777  0.0  45.778013
3   78.192337        9.0  90.968991  39.685558  0.411379  0.0  48.419005
4  102.554043        4.0  44.561913  11.017535  0.447583  1.0  59.161710
```

```
         tau
0   5.658414
1  -0.008397
2   0.785915
3  18.905056
4  23.398746
```

The three panels below are the picture to hold in your head for the rest of the notebook:

1. **The planted effect is spread out** (not a single value) — that spread is the *heterogeneity* that makes targeting possible, and it straddles both 0 and the €8 cost line.
2. **Persuadables cluster in feature space** — high tau (green) lives at particular recency/engagement combinations, which is *why* a model using x can find them.
3. **The confounding, made visible** — the probability of having been emailed *rises with engagement*. Emailed and not-emailed customers are therefore **not comparable as-is**; that is the problem every method below has to overcome.



0.2 3 · Identify — what exactly are we estimating, and when is it even possible?

"Identification" vs "estimation" — the single most important distinction in causal inference. *Identification* asks: *given how the world works, can the effect in principle be recovered from this data at all?* That is a question about **assumptions**, answered before you fit anything. *Estimation* is the statistics you do *once identification is granted.* A beautiful model computed under a false identification assumption is **confidently wrong** — so we keep the two separate.

The estimand (the precise thing we want). We climb a ladder of three related quantities:

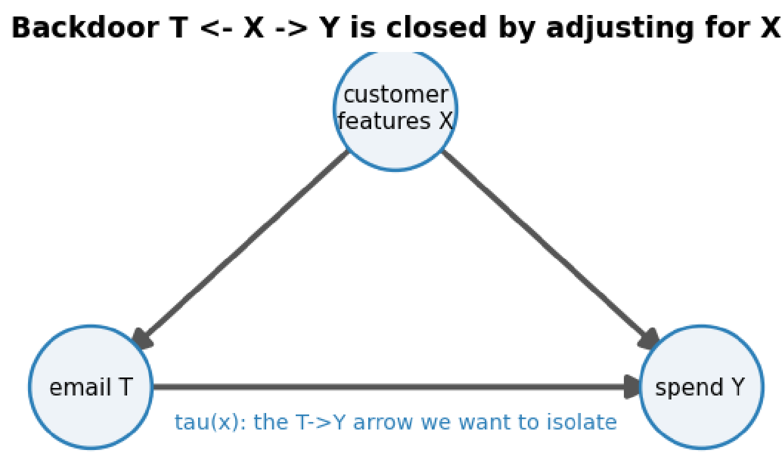
quantity	plain meaning	formula
ATE — Average Treatment Effect	"does emailing lift spend *overall*?"	$\mathbb{E}[Y(1) - Y(0)]$
CATE — Conditional ATE, $\tau(x)$	"the effect *for customers like* x" -> *who* to target	$\mathbb{E}[Y(1) - Y(0) \mid X = x]$
policy value	"the € we make from a given targeting rule"	$\mathbb{E}[(\tau(x) - c)\,\mathbb{1}\{\text{we target } x\}]$

($\mathbb{E}[\cdot]$ is just "average over customers"; $\mathbb{1}\{\cdot\}$ is 1 when the condition holds, else 0.) The uplift decision needs all three, but everything rests on the CATE $\tau(x) = \mu_1(x) - \mu_0(x)$, where $\mu_t(x) = \mathbb{E}[Y \mid T = t, X = x]$ is the *average spend of emailed (t=1) or not-emailed (t=0) customers who look like x.*

The three assumptions that make $\tau(x)$ recoverable (each is revisited in Depth B):

1. **Unconfoundedness** (a.k.a. *no unmeasured confounders*, "ignorability"): once we condition on the measured features X, who got emailed is as-good-as-random. Formally $\{Y(0), Y(1)\} \perp T \mid X$ ($\perp$ reads "is independent of"; $\mid X$ "once we hold the features fixed"). *Guaranteed by design in an A/B test; an assumption otherwise* — and the big one, because it can't be checked from data.
2. **Positivity / overlap**: every kind of customer had *some* chance of being emailed and *some* chance of not — $0 < e(x) < 1$, where $e(x) = P(T{=}1 \mid X{=}x)$ is the **propensity score** (a customer's probability of having been emailed). If some group was *always* emailed, we have no comparison for them and are extrapolating. Checked in Step 4.
3. **SUTVA**: one customer's email doesn't change another's outcome (no word-of-mouth spillover), and there's only one version of "the email."

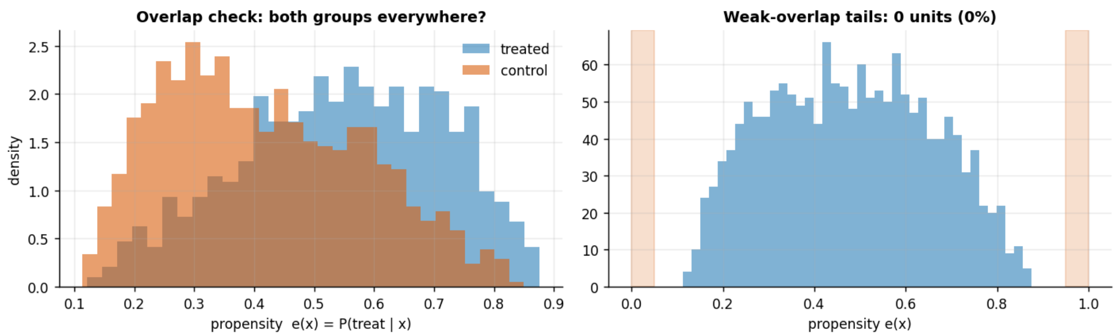
Under 1–3, the **backdoor adjustment** (or *g-formula*) says $\tau(x) = \mu_1(x) - \mu_0(x)$: comparing emailed vs not-emailed *within* look-alike customers removes the confounding. The DAG (directed acyclic graph — a picture of what causes what) shows why. The "backdoor path" email <- features -> spend is the route by which confounding sneaks in; conditioning on the features **blocks** it.



0.3 4 · Estimate — check overlap first, then fit two independent ways

Discipline: look at overlap *before* estimating. Adjustment can only compare emailed vs not-emailed customers where *both* actually exist. If, at some feature combination, everyone was emailed, there is no control to compare against and the model is silently extrapolating. So we first plot the **propensity score** $e(x)$ by arm (emailed vs not); healthy overlap means the two histograms cover the same range, with few customers stuck at 0 or 1.

```
Overlap is adequate - only 0 of 1600 customers fall in the <0.05 / >0.95 tails.
```



Now the estimators. We fit the effect two completely different ways and check they agree — agreement across independent methods is our first credibility signal.

- **BCF — Bayesian Causal Forest** (our primary CATE model). Under the hood it's **BART** (Bayesian Additive Regression Trees — a flexible sum-of-trees model that needs no functional-form guesswork and returns a posterior). BCF is a causal-tuned BART with two tricks: it feeds the estimated propensity $e(x)$ into the model so *targeted-selection confounding* is absorbed (**targeted selection** is our data's specific confounding pattern: marketers emailed customers *because* they looked likely to spend, so the propensity $e(x)$ and the baseline spend $\mu_0(x)$ move together — feeding $\hat{e}(x)$ in lets the model soak up exactly that overlap, which is why BCF beats a plain T-learner precisely here, and only here, per Step 7's caveat), and it puts **fewer trees on the treatment surface** than on the baseline surface (a weak, tree-count form of BCF's shrinkage-toward-homogeneity prior), so the model only reports heterogeneity when the data really support it (rather than reading noise as signal). Output: a posterior array of shape `(draws, customers)` — one plausible $\tau(x)$ curve per posterior draw.
- **AIPW — Augmented Inverse-Propensity Weighting** (a cross-check on the *average*). It's **doubly-robust**: it combines an outcome model (gradient-boosted trees here) and a propensity model, and is correct if *either one* is right. It's a **methodologically different** cross-check (a different outcome model plus explicit propensity weighting), so agreement is reassuring though not fully independent.

The fitted model (BCF), in symbols. With $\hat{e}(x)$ the propensity estimated above:

$$y_i \sim \mathcal{N}(\mu_i,\, \sigma^2), \qquad \mu_i = f_{\text{prog}}(x_i,\, \hat{e}(x_i)) \;+\; f_\tau(x_i)\, T_i, \qquad \sigma \sim \text{HalfNormal}(15),$$
$$f_{\text{prog}} \sim \text{BART}(m \text{ trees}), \qquad f_\tau \sim \text{BART}(m/2 \text{ trees}),$$

where f_{prog} is the prognostic surface (baseline spend as a flexible function of the features *and* the propensity — feeding $\hat{e}$ in is the trick that absorbs targeted-selection confounding) and f_τ is the treatment surface, whose posterior **is** the CATE: $\hat{\tau}(x) = f_\tau(x)$. Halving the tree count on f_τ is the shrink-toward-homogeneity prior from the bullet above, in its concrete form. (The HalfNormal(15) on σ is just a mild prior keeping the noise sd positive and around the tens-of-euros scale — nothing hinges on the exact 15.)

The cross-check (AIPW), in symbols. AIPW averages one *influence-function* score $\hat{\psi}_i$ per customer — think of it as each customer's individual contribution to the average effect, engineered so that errors in either model wash out (see below) — built from an outcome model $\hat{\mu}_t(x)$ (predicted spend under treatment $t \in \{0,1\}$) and the propensity $\hat{e}(x)$:

$$\hat{\psi}_i \;=\; \underbrace{\hat{\mu}_1(x_i) - \hat{\mu}_0(x_i)}_{\text{outcome-model guess}} \;+\; \underbrace{\frac{T_i\,(y_i - \hat{\mu}_1(x_i))}{\hat{e}(x_i)} \;-\; \frac{(1-T_i)\,(y_i - \hat{\mu}_0(x_i))}{1-\hat{e}(x_i)}}_{\text{inverse-propensity-weighted residual corrections}}, \qquad \widehat{\text{ATE}} \;=\; \frac{1}{n}\sum_{i=1}^{n} \hat{\psi}_i.$$

This equation is *why* “doubly robust” is more than a slogan: if the outcome models are right, the residuals $y_i - \hat{\mu}_{T_i}(x_i)$ are mean-zero and the correction terms vanish on average, leaving the correct first term; if instead the propensity is right, the inverse-weighted residuals exactly repair whatever bias the outcome models leave behind. Either model correct suffices — only both wrong breaks the estimator.

The pymc-bart gotcha, baked in. Scoring BART counterfactuals (“what would this customer have spent under the *other* treatment?”) silently returns *frozen training predictions* — every effect exactly 0 — unless the tree node is resampled. `cmp.estimators` always does this correctly, and a unit test asserts non-zero recovered effects, so you never hit the trap.

Did the sampler converge? (the check behind every posterior in this cookbook). All the Bayesian fits here draw posterior samples with **MCMC** (Markov-chain Monte Carlo — the standard algorithm for drawing from a posterior), and a posterior is only trustworthy if the sampler actually converged. Three standard diagnostics, in plain terms:

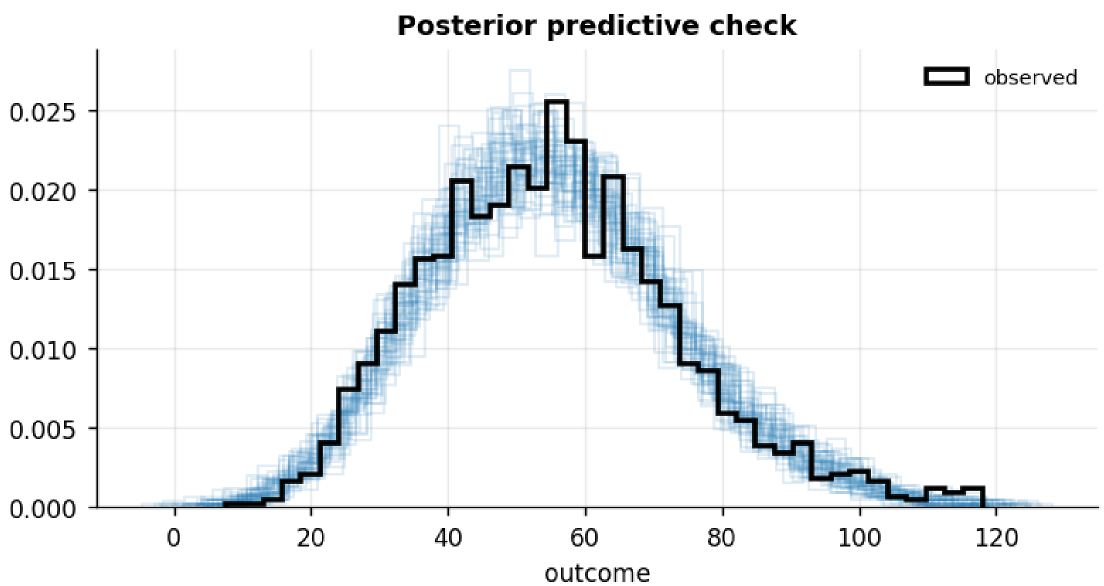
- **R-hat** — run several independent chains and compare them: if they disagree about where the posterior is, they haven’t converged. R-hat ~ 1.00 is good; above ~1.01 means keep sampling (or fix the model). One honest exception, applied in the very next cell: BART is fitted with the particle-Gibbs **PGBART** sampler, and scalars coupled to its tree sums (like the observation-noise sd we monitor) routinely sit at r-hat ~ 1.01–1.02 in FULL runs even when the fit is fine — and, unlike pure NUTS, raising draws grinds that down only slowly. So for the BART-based fits we read **<= 1.01 as a clean pass, 1.01–1.02 as *tolerated*** (with the 8-seed stability sweep in Step 5 as the backstop that the *estimates* are stable), **and > 1.02 as a genuine investigate signal**. The printout below applies exactly these bands, so the verdict you see is computed, not asserted.
- **ESS** (*effective sample size*) — consecutive draws are autocorrelated, so 1,000 draws may carry the information of only 100 independent ones. ESS is that independent-equivalent count; you want it comfortably in the hundreds for stable interval endpoints.
- **Divergences** — a NUTS-specific warning that the sampler hit posterior geometry it couldn’t integrate; more than a handful means the posterior is suspect (raise `target_accept` or reparameterize). BART is fitted with a particle-Gibbs sampler instead of NUTS, so divergences don’t apply here — but R-hat and ESS still do.

PyMC and CausalPy print these warnings automatically when something is off — silence is the pass signal. We keep them out of the notebook read-outs to stay focused on the causal logic, with one exception worth knowing about: notebook 06’s MMM is hard enough geometry that FULL runs need `target_accept = 0.99` and deeper trees to reach R-hat ~ 1.01.

```
BCF   ATE = €6.39   (true €5.95)
BCF   sampling convergence (4 chains): max r-hat 1.010 - min ESS 649 -
divergences 0
      r-hat verdict: PASS - max r-hat 1.010 is at or under the ~1.01 bar
AIPW  ATE = €6.28   90% CI [5.39, 7.16]   (doubly-robust)
BCF's ATE lies within the AIPW 90% CI - two methodologically different
estimators agree, our first credibility check.
```

Model criticism — the posterior predictive check (PPC). Before we *read effects off* a model, we ask whether it can even reproduce the data it was trained on. We draw fake ("replicated") datasets from the fitted model and overlay their spend distribution on the real one. If the real data looked nothing like what the model can generate (wrong spread, missed skew), the effect estimates would be untrustworthy no matter how tight their intervals.

```
Posterior-predictive check: 90% of observed spend falls within the replicates'
5-95% range (~90% if calibrated) - no gross misfit (posterior sigma ~ €9).
```



0.4 5 · Validate — did we recover the truth, is the uncertainty honest, and does it rank well?

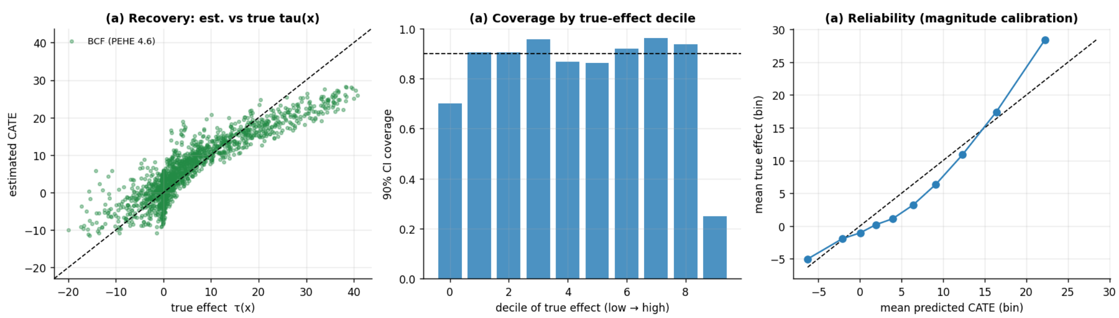
This is the payoff of simulating a known truth. Three questions, each with its own diagnostic:

(a) Recovery & calibration — are the numbers right, and honestly uncertain? - PEHE (*Precision in Estimation of Heterogeneous Effects*) — the root-mean-square error between the estimated $\hat{\tau}(x)$ and the true $\tau(x)$, in €. Lower is better; it's the CATE analogue of RMSE. - **Coverage** — a *90% credible interval* should contain the true effect about 90% of the time. We also break coverage out *by decile of the true effect* to catch a model that's well-calibrated on average but overconfident for the biggest effects. - **Reliability** — bucket customers by predicted effect and plot mean-predicted vs mean-true; on the 45° line means the *magnitudes* are calibrated, not just the ranking. - **Sharpness** — the average width of the intervals. Sharp *and* calibrated is the goal; sharp *and* miscalibrated is just overconfidence, which is why we never report sharpness alone.

(b) Ranking quality — for targeting, the *order* matters more than exact magnitudes. - **Qini / uplift curve** — walk down customers from highest to lowest predicted effect, contacting as you go, and plot the cumulative *true* effect captured. A good model's curve bows up toward the **oracle** (a hypothetical model that knows every true tau) and well above the **random** diagonal. - **AUUC** (*Area Under the Uplift Curve*) — that curve's area, normalised so **1.0 = oracle-perfect ranking** and **0 = no better than random**. The single number for "how well do we rank?" - **Uplift-by-decile** — average *true* effect within each tenth of predicted effect; a good model shows a clean descending staircase (top decile = biggest real effect).

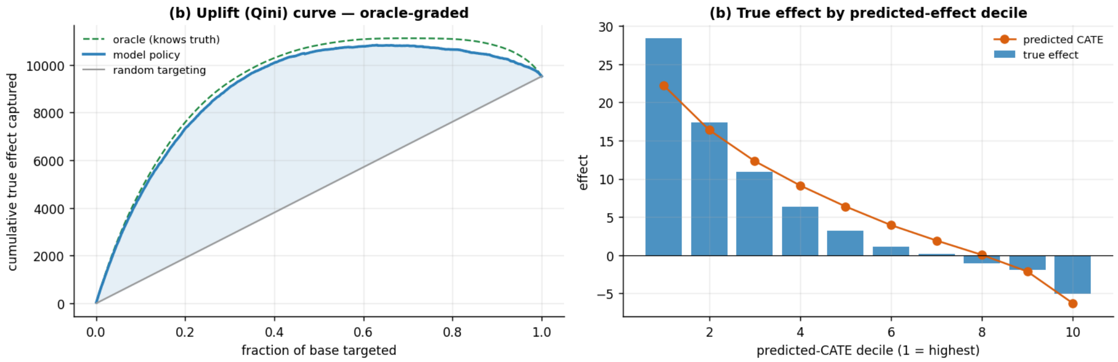
(c) Stability — is the result a fluke of one random draw? We refit on several fresh simulations and check the recovery is consistent.

```
PEHE €4.63  ·  corr(est,true) 0.91  ·  90% coverage 83%  ·  interval sharpness
€11.3
```



Read-out. The recovery scatter and the line just printed tell one story: a high est-vs-true correlation and a small **PEHE** (a few € against true effects that span roughly −€10 to +€40) say the *ranking* is strong and the magnitudes are close, while the printed **90% coverage** lands a touch under its 90% target — a mild over-confidence that the middle panel pins to the largest-effect deciles and that Step 5's coverage caveat (below) returns to. Sharpness is read only alongside that coverage, never on its own, so a narrow-but-miscalibrated interval can't pass itself off as precision.

```
AUUC = 0.94 of oracle (1.0 = perfect ranking, 0 = random).
(oracle-graded: uses the known tau; on real RCT data use metrics.qini_observed,
which reweights
 observed outcomes by arm and needs no counterfactual - see §6b.)
Top decile true effect €28.4  vs  bottom decile €-5.0 - the model concentrates
the real uplift where we'd actually spend.
```



Heads-up on the output below: the sweep deliberately refits BCF with **short 80-draw chains** — enough for stable posterior *means* (all we read here), not for interval-grade diagnostics. The repeated `Only 80 samples per chain` notices are that choice talking; we leave them visible rather than model the habit of silencing sampler diagnostics on a cell whose output we quote.

```
(c) Stability across fresh simulations:
 seed  ATE_hat  true_ATE  PEHE  AUUC
  100     5.92      5.61  5.26  0.92
  101     5.82      6.12  5.10  0.95
  102     5.68      5.64  5.02  0.95
  103     6.19      5.63  4.81  0.94
  104     6.05      5.75  5.29  0.93
  105     5.87      5.78  5.25  0.92
  106     5.18      5.34  4.75  0.95
  107     6.29      5.84  5.04  0.93

ATE recovery bias across seeds: +0.16 (sd 0.30) - consistent, not a lucky seed.
```

0.5 6 · Decide, in euros

An effect estimate is useless until it becomes an action and a number a manager signs off on. Three moves:

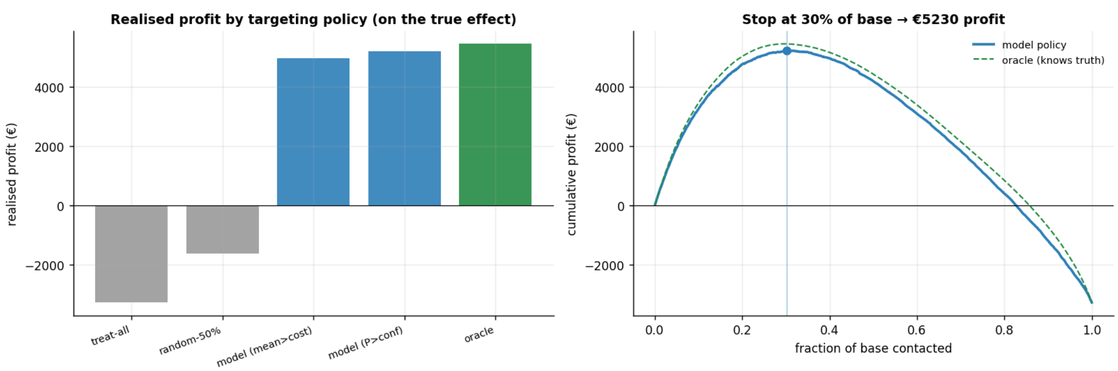
Compare targeting policies on realised profit. We evaluate five rules on the *known truth* (only possible in simulation) so we can see, in €, how much targeting beats the alternatives and how close it gets to the unbeatable oracle: - *treat-all* (email everyone), *random-50%*, - *model (mean > cost)* — email where the posterior-mean effect beats the €8 cost, - *model (P > conf)* — the **honest rule**: email only where we're *confident* the effect beats cost (see below), and - *oracle* — email exactly the customers whose *true* effect beats cost (the best any model could do).

The honest targeting rule. We do **not** target on "mean effect > cost." A customer whose point estimate says "target" but whose interval straddles the cost line is a coin-flip, not a yes. So we target where $P(\tau(x) > \text{cost})$ is **high** (here > 0.8) — probability, from the posterior, that the effect really clears the cost.

The profit curve. Rank customers by estimated effect, contact down the list, and plot cumulative (true effect - cost). It rises while we're adding profitable customers and falls once we hit unprofitable ones; its **peak is the optimal stopping point** — "email the top X%, make €Y."

policy	profit	frac_contacted	profit_vs_all
treat-all	-3277	100%	0
random-50%	-1633	49%	1644
model (mean>cost)	4981	40%	8258
model (P>conf)	5204	29%	8481
oracle	5459	30%	8735

```
Email everyone: €-3,277  ·  honest model rule: €5,204 (95% of the oracle's
profit), targeting 458 of 1600 customers.
```



0.6 6b · The same pipeline on real data (Hillstrom email experiment)

Everything above used *simulated* data so we could grade recovery. Here we run the identical estimator on a **real, public dataset you don't have to provide** — it's fetched over the network from a public URL:

> **Hillstrom / MineThatData (2008)** — 64,000 real customers randomly assigned to *Men's email*, *Women's email*, or *no email*, with downstream visits, conversion and spend. We reduce it to a binary *Women's-email vs no-email* comparison. Because assignment was **randomized (a true A/B test)**, the average effect is trustworthy with no adjustment — but there is **no ground-truth per-customer effect** here (real customers only live one future), so we can estimate and decide, but we *cannot* grade CATE recovery. That's exactly what production looks like: the validation lives in the simulation above; the money is made here.

The cell is **gated** — it fetches only if you set `CMP_REAL=1` (so the offline test suite stays deterministic). Set it and re-run to see the real-data version.

> **What you'll see — a recorded run.** The committed notebook keeps the gate off so the repo runs offline, but for reference a recorded run of this cell (FULL profile, `CMP_REAL=1`) gives: the two-arm slice has **n = 42,693** customers (21,387 emailed, 21,306 held out), and the randomized comparison yields a causal **ATE ~ €0.42 per customer (SE ~ €0.13)** — a genuine, significant lift that pays for itself at any realistic

cost of sending an email. On the 4,000-customer subsample, BCF flagged roughly one customer in ten as a confident target (P(effect>0) > 0.8), and ranking by predicted uplift put on the order of **€0.9k of observable incremental spend in the top-30% bucket versus essentially nothing (slightly negative — spend is noisy and zero-inflated) for a random 30%** — a real observable-Qini win, computed with no counterfactual. Two different kinds of number in that list: the sample sizes and the ATE are exact properties of the fixed public dataset and reproduce to the cent; the BCF-derived read-outs (the ~10%, the Qini split) are BART-posterior quantities and pymc-bart is not seed-stable across machines, so treat those as orders of magnitude and run `CMP_REAL=1` to get your own.

```
Real-data section skipped. Set CMP_REAL=1 and re-run this cell to fetch the
Hillstrom experiment.
On real data there is no ground-truth tau, so Steps 5's recovery checks do not
apply.
```

0.7 7 · Caveats — the honest failure modes

- **Unconfoundedness is untestable.** If a *hidden* driver moved both who got emailed and how much they spend (say a salesperson's hunch we never recorded), the estimate drifts. Depth B quantifies exactly how strong such a confounder would have to be to flip the decision — as an **E-value**.
- **Overlap matters.** Where some kind of customer was *always* emailed, we're extrapolating, not comparing. We checked the tails in Step 4 and can trim them.
- **Estimator choice is not cosmetic.** The S-learner (Depth A) regularises the treatment signal away and under-detects heterogeneity — it fails on both PEHE *and* ranking (AUUC).
- **BCF ~ T-learner when confounders are fully observed.** The propensity trick earns its keep specifically under *targeted selection*; don't oversell it as free lunch.
- **Sleeping dogs are real.** Negative-effect customers mean "email everyone" can be *value-destroying*, not merely wasteful — which is exactly what the policy comparison in Step 6 showed.

---

1 Depth A · Estimator bake-off & failure modes

Why a whole section on picking the estimator? Because the choice is not cosmetic — a popular default (the S-learner) systematically **flattens** heterogeneity and would quietly wreck the targeting. We line up three estimators against the known truth, in two worlds, on the metrics that actually matter for targeting.

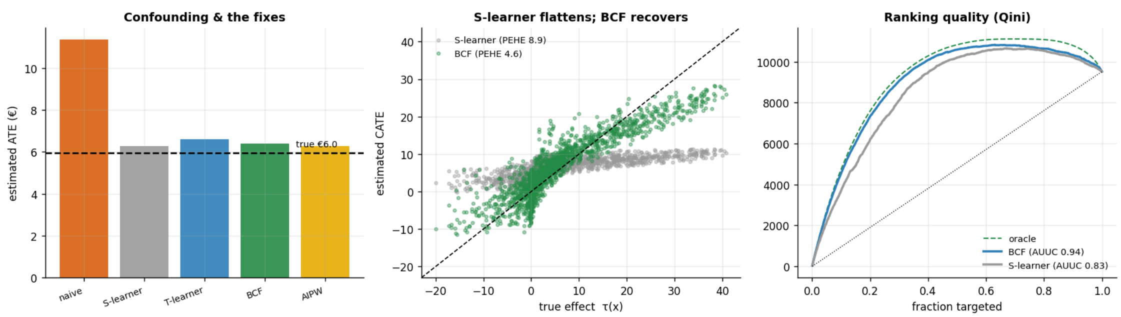
The three estimators (meta-learners — recipes that turn any regression into a CATE estimator): - **S-learner** ("S" = single) — fit *one* model of spend on features *and* treatment together, then read the effect as (prediction with T=1) − (prediction with T=0). Simple, but the model can shrink the treatment's influence toward zero, **flattening** real differences. This is the labelled failure mode. - **T-learner** ("T" = two) — fit *separate* models on the emailed and not-emailed customers, then subtract. A solid default when treatment is randomized. - **BCF** — the propensity-aware, shrink-to-homogeneity model from Step 4; best under confounding.

The two worlds: - **Randomized** — treatment was a coin flip, so *unconfoundedness holds by* *design* and even simple methods are unbiased on the average. - **Observational** — the confounded world we've been in; here the naive "emailed − not-emailed" average is badly biased, and the estimator has to actively undo the confounding.

The metrics: PEHE (magnitude error, €), corr (ranking correlation), **AUUC** (ranking quality vs oracle), ATE_bias (average-effect error, €), and cov90 (are the 90% intervals honest?).

```
Naive 'emailed - not emailed' (observational): €11.4 vs true €6.0 -> bias +5.4
(confounding nearly doubles the apparent effect)
```

	regime	estimator	PEHE	corr	AUUC	ATE_bias	cov90
0	randomized	S-learner	8.83	0.76	0.85	-0.01	0.40
1	randomized	T-learner	4.17	0.93	0.96	-0.59	0.89
2	randomized	BCF	4.67	0.91	0.95	-0.49	0.87
3	observational	S-learner	8.93	0.76	0.83	0.32	0.36
4	observational	T-learner	4.25	0.92	0.93	0.64	0.91
5	observational	BCF	4.87	0.90	0.93	0.33	0.85



Read-out. Two robust messages and one honest caveat — read them off the table above. (1) **The S-learner is the clear loser** in both worlds on the metrics that matter for targeting: highest PEHE, lowest AUUC, and badly under-covering intervals (`cov90` ~ 0.4). Its single shared surface regularises the treatment signal toward zero, so it **flattens** heterogeneity even with no confounding to fight — which wrecks the per-customer ranking. Tellingly, though, its *average* effect is still fine: in the *randomized* world its ATE bias is ~ 0 (in fact the smallest in the table). Flattening hurts the per-customer effects and their ranking, not the mean — which is exactly why a response-style average would miss the problem and only the heterogeneity metrics expose it. (The T-learner's and BCF's randomized ATE biases, ~ −0.6 and −0.5, are themselves small against the ~€6 effect, so all three recover the *level*; the S-learner's fault is heterogeneity, not the average.) (2) **The T-learner and BCF land together** — near-identical PEHE, correlation and AUUC — and both agree with the methodologically different, doubly-robust AIPW estimator (a boosted-tree cross-check) on the ATE, our cross-method credibility check. *Which* of the two edges the other on any single metric **wobbles run-to-run in FAST mode** (few draws and trees), so we don't over-read a small PEHE gap; with fully observed confounders and good overlap the two are *expected* to coincide (Step 7's caveat), and BCF stays the primary model because its propensity input earns its keep under **targeted selection**, the regime Depth B stress-tests.

Coverage — the honest caveat we don't hide. BCF's 90% intervals tend to

under-cover on the largest true effects — read the `90% coverage` printed above (and the bake-off `cov90` column) for *this* run, and see the coverage-by-decile panel in Step 5, where the top true-effect decile is worst. The exact figure **wobbles run-to-run** because pymc-bart's sampler is not seed-reproducible; the T-learner's intervals are usually a touch better calibrated, and the S-learner under-covers badly (the labelled failure-mode). The takeaway is deliberately conservative: heterogeneity is hardest to pin exactly where the effect is biggest, so we treat every `P(tau>cost)` downstream as a decision *input*, not gospel, and keep the "test the straddlers" recommendation — it hedges precisely where calibration is thinnest.

---

2 Depth B · Identification rigour & sensitivity

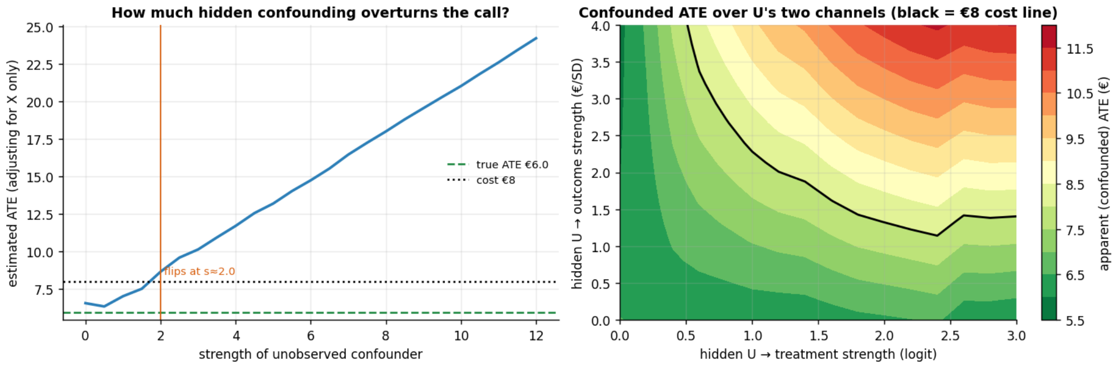
We already confirmed **overlap** (Step 4). The deeper worry is **unconfoundedness**, which is *untestable* — so the honest question is not "is it true?" but **"how fragile is our conclusion to a confounder we didn't measure?"**

We simulate a hidden driver U that moves *both* who gets emailed and how much they spend, sweep its strength, and refit adjusting for the observed features **only** (blind to U, as we'd be in reality). We report two complementary things a CMO can act on:

- the **tipping point** — the confounder strength at which the (wrongly) adjusted effect crosses the €8 cost line and would flip us from "target selectively" to "email everyone pays" (a *decision*-flip number);
- the **E-value** (VanderWeele-Ding) — the minimum strength of association a hidden confounder would need with *both* treatment and outcome, on a **risk-ratio scale** — a risk ratio being how many times more likely an outcome is under one condition than another, i.e. a multiplicative scale — to explain the *positive effect* away entirely (to zero). The E-value is defined only on the risk-ratio scale, so we standardize the euro effect by the outcome SD (VanderWeele's continuous-outcome rule) and compute it on the **AIPW estimate** — never on the simulation truth, which a real analyst never has. A large E-value means "you'd need an implausibly strong hidden confounder to overturn this"; a modest one (as we find here) means the robustness is real but not decisive.

The right panel makes the tipping point two-dimensional: it is a **genuinely refit** surface of the confounded ATE as U's two channels (into treatment, into outcome) vary independently — so the black cost line traces the real (a, b) frontier, not a drawn-in gradient — omitted-variable bias (**OVB**) in two dimensions, where the bias scales with *both* channels, $a \times b$.

```
Tipping point: the observed-feature-adjusted ATE crosses the €8 cost line at
hidden-confounder
strength ~ 2.0 - beyond that, confounding alone could fake an 'email everyone
pays' signal.
E-value 2.07 (at the near CI limit, 1.94): to explain the positive €6.28 effect
away to zero, a hidden confounder would need ~2.1x association (risk-ratio
scale) with BOTH email
and spend. That is moderate, not decisive - ask a domain expert whether anything
that strong went unmeasured.
```



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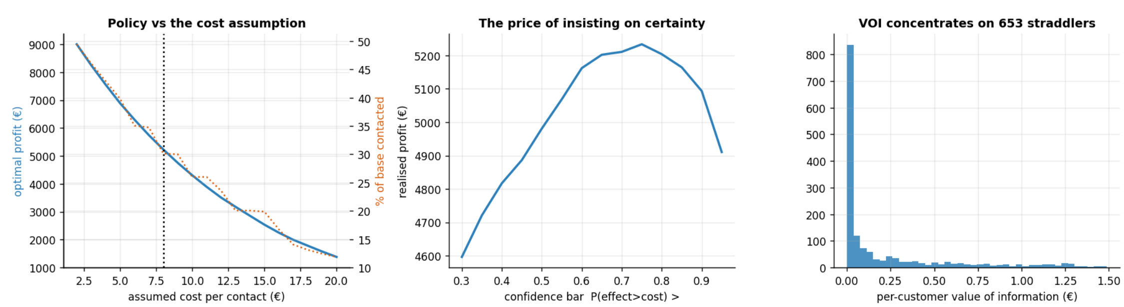
3 Depth C · Euro policy — sweeps, value of information & test design

The decision is a number *under assumptions*. We stress the two a manager will argue about — the per-contact **cost** and how **confident** we insist on being — and then price the remaining uncertainty and turn it into a concrete experiment.

Value of information (VOI). Uncertainty has a *price*: the euros we lose by acting on the noisy estimate instead of the (unknown) truth. It concentrates on the **straddlers** — customers whose credible interval crosses the cost line, so we genuinely can't tell if they're worth contacting. The total VOI on the straddlers is an **upper bound** on what any experiment on them can be worth: it prices *perfect* information, whereas a finite A/B test resolves only part of the uncertainty and so is worth strictly less. Still, it sets the ceiling on sensible test spend — and tells you on whom.

```
Total value of information €351; €340 of it sits on the 653 'straddler'
customers (41% of the base) whose interval crosses the cost line.
```



3.0.1 The one-paragraph decision

> Targeting the model's *confident* persuadables makes materially more money than emailing everyone (which here **loses** money to sleeping dogs), and captures most of the ora-

cle's profit. The call is robust to plausible cost assumptions; hidden confounding would have to be *moderately* strong to overturn it — the confounded ATE only crosses the cost line at confounder strength ~ 2.0, and the positive effect carries an E-value ~ 2 (real but not decisive robustness, so this is a finding to keep watching, not to bank). The uncertainty that remains is worth real money and is concentrated on the "straddler" customers whose interval crosses the cost line — **so the follow-up is a small randomized A/B test on the straddlers, not a blanket send.** Act where we're sure; measure where we're not. *(One honest caveat: BCF's per-customer intervals undercover the largest individual effects — the coverage-by-decile diagnostic above shows the top decile falling short of nominal — so the straddler set is a conservative upper bound, and the biggest-effect customers' tau_hat should be read as a floor, not a point.)*


```
{"true_ate": 5.952, "bcf_ate": 6.386, "aipw_ate": 6.283, "PEHE": 4.633,
"AUUC": 0.937, "coverage90": 0.8269, "profit_treat_all": -3277.0,
"profit_model": 5204.0, "profit_oracle": 5459.0, "sensitivity_tip": 2.0,
"e_value": 2.07, "VOI_total": 350.6, "n_straddlers_to_test": 653}
```